

Your Title in Title Caps

by

First and Last Name

Under the Direction of Committee Chair, PhD

A Dissertation Submitted in Partial Fulfillment of the Requirements for the Degree of

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in the College of Arts and Sciences

Georgia State University

YEAR

ABSTRACT

Your abstract must be in place at the time of initial submission. In MS theses, the abstract cannot exceed 150-words. In doctoral dissertations, the abstract cannot exceed 350-words.

INDEX WORDS: Sample index words, Sample, Sample, Sample

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DEDICATION

This page is optional but highly recommended. The dedication is usually for acknowledging family or friends and is typically a matter of politeness.

ACKNOWLEDGEMENTS

Again, this page is optional but also highly recommended. You may use this section to express acknowledgement of those who have helped you with this document (your committee members and instructors) and your academic career. This section should be double spaced.

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LIST OF ABBREVIATIONS

ANOVA	analysis of variance.
AP	action potential (“spike”).
CSD	current source density.
ERP	event-related potential.
IACUC	Institutional Animal Care and Use Committee.
LoA	list of abbreviations.
MK-801	an NMDAR pore blocker (“Dizocilpine”; C ₁₆ H ₁₅ N).
NMDA	N-methyl-D-aspartate.
NMDAR	NMDA receptor.
NRH	NMDAR hypofunction.
PSTH	peri-stimulus time histogram.
ROI	region of interest.

PREFACE

The preface is optional. If you choose not to use a preface, just delete the `preface.tex` file in `Frontmatter/`.

CHAPTER 1

GETTING STARTED

1.1 The First Section of the First Chapter

You should probably introduce some stuff here.

1.2 Template structure

This template is a multi-file project, structured in a hierarchical way that allows you to develop and version control different sections of your document in isolation while still being able to compile the whole thing together. The overall typesetting layout of this document (e.g. Frontmatter and main Chapter pages) is defined by `gsudiss.cls` in the Styles directory. The main file (`main.tex`) initializes packages and environments, specifies parameters for citations, linking, and naming conventions, and acts as the master document that includes front, main, and backmatter sections. As written, each chapter is itself a parent file that is included in `main.tex` via the `subfiles` package, and which includes individual sections as separate files, all of which can be compiled as standalone files to reduce time.

This structure can be visualized as:

```
root/
├── main.tex
├── Chapters/
│   ├── ch_01/
│   │   ├── ch01-section.tex
│   │   └── Chapter_01.tex
│   └── ch_xx/
├── Plots/
└── Tables/
```

Given dissertation chapter sections (e.g. Methods, Results) can be quite long, this separation can be beneficial for troubleshooting when errors occur and reduce compilation time when working on targeted sections. Though you could just as easily put all of the content for a chapter in one file.

1.3 Document interactions

Multi-file projects are good for version control and isolated work, but involve navigating back forth between different documents. In VS Code, the editor this template was primarily developed with, you can drag the tab of the open PDF off to side of your window to separate the pane for document viewing alongside its open source tab. It's useful to know how to quickly jump from PDF to its generating `.tex` source code (Reverse SyncTeX), and from source code to its location in the PDF (Forward SyncTeX).

1.3.1 Reverse SyncTeX

`Ctrl + Double-click` on the PDF will snap to the approximate position in the source code file.

1.3.2 Forward SyncTeX

`Ctrl+Alt+J` from your cursor position in the source code will snap to its corresponding location rendered in the PDF.

CHAPTER 2

FORMATTING TEXT

2.1 Section structure

2.1.1 Subsections

Subsections at this depth (`\subsection{}`) will be displayed in the Table of Contents.

2.1.1.1 subsubsection

Though sections at this depth (`\subsubsection{}`) will not be included in the Table of Contents.

2.1.2 Linking and referencing

2.1.2.1 Chapters

The `cleveref` package allows you to reference chapters using a label that you assign to the Chapter name. For example Chapter 1 includes some info on document interactions. Notice that the “1” to the right of “Chapter” is a link that will take you to the beginning of the chapter on Getting Started.

2.1.2.2 Sections

You can reference sections using a label assigned in the same way, and in all cases use both numbers and names. For example, Section structure, assigned here the number Section 2.1, shows you section and subsection depth. Section 2.2.1, Abbreviations, below includes another kind of linking: terms which snap to the list of abbreviations (LoA) page in your frontmatter.

2.2 Content examples

2.2.1 *Abbreviations*

The `glossaries` and `glossaries-extra` packages help you manage the insertion of abbreviations, applied programmatically when inserted with the commands below. Take a look at the `abbreviations.tex` file to see how they're defined. Review the documentation for exhaustive command list.

```
\gls{erp}      % first instance expanded
\gls{erp}      % second + thereafter truncated
\glspl{erp}    % plural
\Acrfull{erp}  % force full expansion, capitalize
```

Produces: (i) event-related potential (ERP), (ii) ERP, (iii) ERPs, and (iv) Event-related potential (ERP).

2.2.1.1 *Compositionality*

The original implementation of this template utilized the `acronym` package, though the `glossaries` package allows more flexibility and control over more complicated cases. For example, `glossaries` by default sorts your defined terms alphanumerically before rendering them in your LoA page. This is more ideal for compositionality in your abbreviations (i.e. using small acronyms as building blocks for others). See the code in `abbreviations.tex` that generates the following example case and their appearance in the LoA:

“N-methyl-D-aspartate (NMDA) elicits responses from NMDA receptors (NMDARs). A variety of neuropsychiatric disease states are associated with NMDAR hypofunction (NRH).”

2.2.1.2 *Pluralization*

Sometimes pluralization of acronyms can safely be achieved by appending an “s” to the end of the final word. This is true in the case of event-related potentials (ERPs). Though other times this pluralization belongs on a different part of the acronym. For example, regions of interest (ROIs) more naturally has its first word pluralized. This behavior can be specified by supplying this to the acronym definition in `abbreviations.tex`.

2.2.1.3 *Alternate forms*

Another noteworthy feature of `glossaries` is the ability to define alternate versions of your acronyms for terms used interchangeably. This is done using wrapper commands that check for user-defined inputs in definitions. For example, in neuroscience:

“action potentials (APs) emitted by a brain cell are often referred to as spikes.”

Both of these commands will link back to the definition of APs in the LoA. This is one of a variety of ways you could possibly implement and customize your management of abbreviations.

Notice also that the definition for MK-801 (an NMDAR pore blocker) includes the drug common name and chemical formula in a description field, which, here, *will not* behave as an alternate form that we can reference using the `altgls{}` form command. This can be useful when you want to supply supplementary information not intended to be used or referenced back to the LoA page from the text. All of these commands are custom, and can be further modified or built upon to meet your needs.

2.2.1.4 Link control

Lastly, you may not want to link every instance of an acronym to the LoA page. You can control this with the starred versions of the commands, which will still apply the same formatting and expansion rules as their non-starred counterparts. For example, an instance of NMDAR using `\gls*{nmdar}` should have no link.

2.2.2 Citations

Here's an example citation of a random paper¹. Notice that the superscripted number is a link that will take you to the corresponding entry in the bibliography. A custom command `\authoryr{}` is also defined for inline citations, as suggested by Aarts *et al.* (2014). Like `glossaries`, the starred version `\authoryr*{key}` is unlinked: Aarts *et al.* (2014). Examine `bibliography.bib` to see what information your bibliography `.bib` file should contain. These can be generated by most modern reference managers, for example Zotero or EndNote.

Important: The citation style chosen here is common in fields like neuroscience, and passed formatting review for my committee/department, though you should check with your advisor or department for the preferred style in your field and adjust the bibliography style in `main.tex` accordingly. Also note that in this template, individual entries of the bibliography are single-spaced, but double-spaced between entries. This behavior can be modified by commands in lines 12–15 of `backmatter.tex`.

2.2.3 *SI units*

The `siunitx` package allows us to insert units in a typographically consistent way, ensuring proper height, spacing, and formatting of units (e.g. prevention of linebreaks). It’s desirable in this context for much the same reasons as `glossaries`.

“The concentration of drug was 0.02 mol/L (20 mM), which can also be formatted as $0.02 \frac{\text{mol}}{\text{L}}$. After a 30 min period for pharmacological action, mice were placed in a chamber, anesthetized under 3 % isoflurane, then fixed in a nose cone delivering the same anesthetic mixture. Samples were extracted, tin foil wrapped, and placed in a -80°C freezer for 25 minutes to achieve firmness.”

Table 2.1 below shows common examples of the displayed output and generating source code for ranges of numbers and scientific units.

Table 2.1: Unit table showing various hyphenated ranges. Shown here are a unitless range, a pre-configured shorthand command for milliseconds, proper display of units involving greek letters, two different commands for the same unit, and lastly two different notations for conveying ratios.

Display	Unit	Source Code
10–20	—	<code>\numrange[range-phrase = --]{10}{20}</code>
100–200 ms	milliseconds	<code>\qtyrange[range-units=single,range-phrase=--]{100}{200}{\ms}</code>
50–100 μm	micrometers	<code>\qtyrange[range-units=single,range-phrase=--]{50}{100}{\um}</code>
2–101 Hz	hertz	<code>\qtyrange[range-units=single,range-phrase=--]{2}{101}{\hertz}</code>
2–101 Hz	hertz	<code>\qtyrange[range-units=single,range-phrase=--]{2}{101}{\Hz}</code>
0.1–0.2 mol L ^{−1}	mol per L	<code>\qtyrange[range-units=single,range-phrase=--]{0.1}{0.2}{\mol\per\L}</code>
0.1–0.2 mol/L	mol per L	<code>\qtyrange[range-units=single,range-phrase=--]{0.1}{0.2}{\mol/\L}</code>

CHAPTER 3

FIGURES & TABLES

3.1 Visualizing data

Tables, equations, and figures are the primary ways of communicating data or theory. They can also be linked and referenced using the same `\label{}` scheme as chapters and sections, which is especially useful for large documents where the order of presentation may change during writing and editing. Much like the way modern reference managers handle citations, this makes the numbers or letters assigned to these elements dynamic and robust to changes in the document. Depending on the editor you use, these unique label namespaces are also *remembered*, allowing you to tab complete them.

3.1.1 Tables

3.1.1.1 Formatting control

Above is an example table that demonstrates control of column width and alignment, missing value handling, font size and weight, row spacing, and the usage of horizontal rules (see Table 2.1). Table captions are placed above figures in CAS dissertations. The lines of source code shown in Table 2.1 are aliased inside of the `.vscode` directory in the `diss-latex.code-snippets` file, allowing you to type the prefix—e.g. `qrh` (“quantity range hyphenated”) for `qtyrange`—and tab its completion, followed by tabbing your cursor through the three inputs.

3.1.1.2 Standalone compilation

For a large dissertation document, troubleshooting table formatting can take a long time between edits due to increasing document compilation time. This template includes a **standalone** package that allows you to compile individual `table.tex` files in isolation, which can help more rapidly prototype your design. The `table-testing.tex` file in the `Tables/` directory includes the requisite document class and example code to do this. You can replace its contents with the code from your table files and then copy the final version back into the main document. It's also possible to create a snippet that inserts this boilerplate standalone document class routine inside of a blank source `.tex` file.

3.1.1.3 Page orientation

Table 3.1 below shows an example of a table positioned on its own page in landscape orientation.

Table 3.1: Example of a page with landscape orientation for wide tables.

Blah	Blah	Blah
-------------	-------------	-------------

Stuff	Things	etc.
...	...	...

3.1.2 Equations

Below is an example mathematical equation that incorporates more units and abbreviations, as well as a demonstration of typesetting inline math.

Current source density (CSD), Φ , was calculated by a numerical approximation of the second spatial derivative of voltage, expressed as:

$$\Phi_{t,c} = -\frac{\phi_{t,c+1} - 2\phi_{t,c} + \phi_{t,c-1}}{\Delta c^2 \times g} \quad (3.1)$$

where $\phi_{t,c}$ represents recorded voltages at time-point t and channel c on the linear electrode array, such that constant Δc is the spacing between adjacent contacts. The constant g in Equation (3.1) represents the conductivity of the extracellular medium, defined for these analyses as 0.3 S/m, consistent with putative bounds of extracellular impedance measurements ($\sim 0.3\text{--}0.6 \text{ S m}^{-1}$) for frequencies below 10 kHz.

Notice how `\siemens\per\m`—much like the examples in the above table—can be displayed as a fraction or exponential notation. Also note that equations are unique in that they are numbered inside of a parenthetical when referenced, as is the convention for many fields. Lastly, the above equation can be shortened when you reference it with its lowercase `\cref{eq:csd}` command, which appears as eq. (3.1). Parenthetical or casing behavior can likely be redefined to the conventions of your field and dissertation guidelines, such as by substituting the default parentheses for brackets or always capitalizing a shorthand “Eq”.

3.1.3 Figures

Figures are displayed with `\includegraphics{}` commands inside of figure environments, which allow you to control their placement and formatting. They can be anything from `jpegs`, `pngs`, and `tiffs` to vector graphics like `pdfs`. Figure 3.1 below shows an example of a single image. We can also make an abbreviation for this figure reference. For example, `\fshort{}`, defined in the `usercommands.tex`, allows specification of the prefix:

“Fig. 3.1 below shows an example of a single image.”

3.1.3.1 A single image



Figure 3.1: An individual figure!

3.1.3.2 Multi-panel figures

Figures with multiple panels are a more interesting case. This template includes functionality that allows creation of in-text links to individual subpanel captions and programmatic formatting of the prefix inside of the main figure caption. Here, we specify a parenthetical style with an inset uppercase letter (see the caption of Figure 3.2). Notice in the source code how panel letters are not hard coded, but instead assigned and incremented by the number and order of `\subref{}` commands.

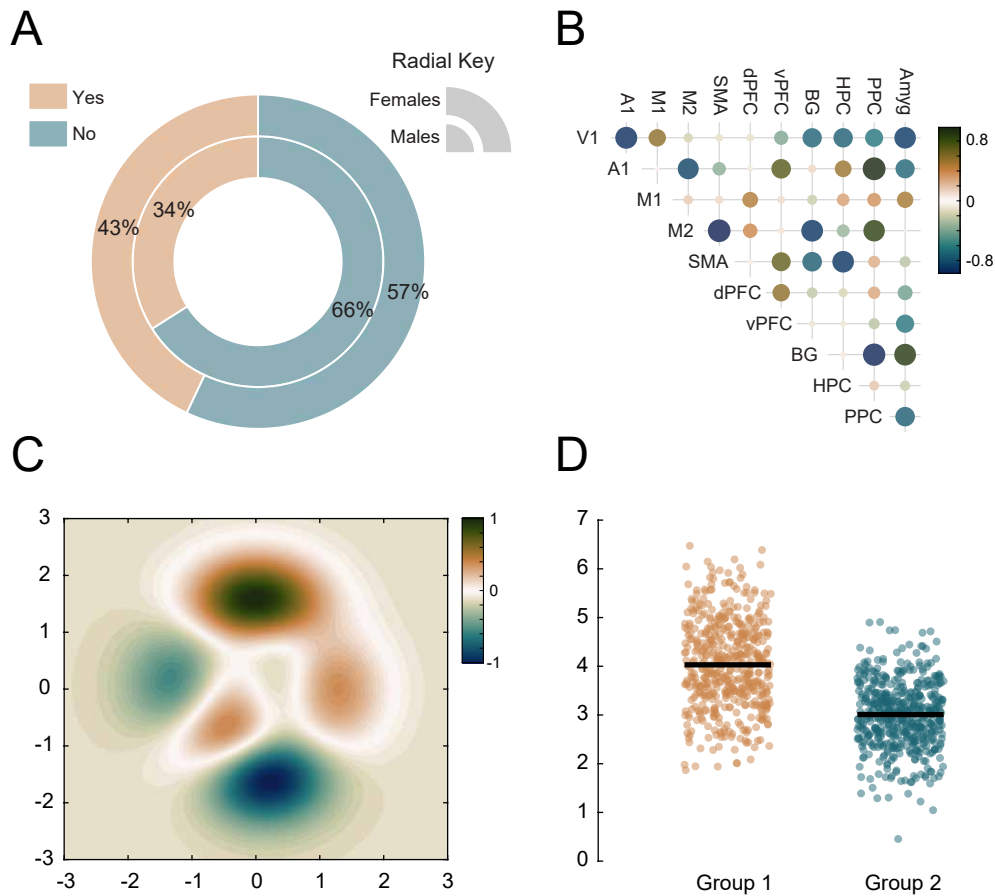


Figure 3.2: Example of a figure with subpanels. (A) Donut plot of the proportions of male and female survey responses. (B) Correlation matrix of activity between brain regions. (C) Typical MATLAB contour plot. (D) Distributions of data for two groups.

Linked references between figures (e.g. Figures 3.1–3.2) and subpanels of the same figure (e.g. Fig. 3.2A–D) are enabled via custom additions to `usercommands.tex`. These reference the subcaption label by their namespace used in `\subref{}`, and will automatically update when you add or remove figures and panels, making them robust to new additions. You can modify these commands to print short or long forms of the “figure” prefix, or to change the style of the range conjunctions. These custom commands for ranged references can *themselves* be aliased in the `diss-latex.code-snippets` file, allowing you to tab complete their insertion by only keying in a few letters of the prefix. Here, they are `frng` and `spr`, respectively.

Figures in this example document are included as a single file regardless if its individual or multi-panel. If you needed to generate your figure panels individually inside of $\text{\LaTeX}$, the `subcaption` package should provide a `subfigure` environment enabling this. Though this is not documented here, as post-processing of figures exported from computing languages like Python, R, or MATLAB inside of vector graphics softwares is more common in scientific applications.

Section A.2 below describes how Appendix Figures and Tables fit into the chapter-figure number notation seen in the Table of Contents and List of Figures

1. Aarts, E., Verhage, M., Veenliet, J. V., Dolan, C. V. & Van Der Sluis, S. A solution to dependency: Using multilevel analysis to accommodate nested data. *Nature Neuroscience* **17**, 491–496 (2014).

A Chapter 3 supplementary material

A.1 Supplemental information

Appendices in this template are typeset single-spaced by default. This can be changed in `backmatter.tex`, though consult with formatting review and the guidelines for your department. There's a variety of different ways you can structure an appendix based on your field. In science, these might be the supplementary materials that accompany individual chapters or published manuscripts. In other fields, they may correspond to derivations, proofs, algorithms, equations, methods, analyses, protocols, or attempted methodologies not germane to the main body text.

A.2 Supplemental figures

Notice that all figures in this template are enumerated first by the chapter number, followed by a second number representing the order in which it appears in the chapter. For Appendices, this template correctly enumerates its encompassed figures and tables by its letter rather than an additional chapter number (see Figure A1). The same logic applies for subpanel references, such as Fig. A1A–B.

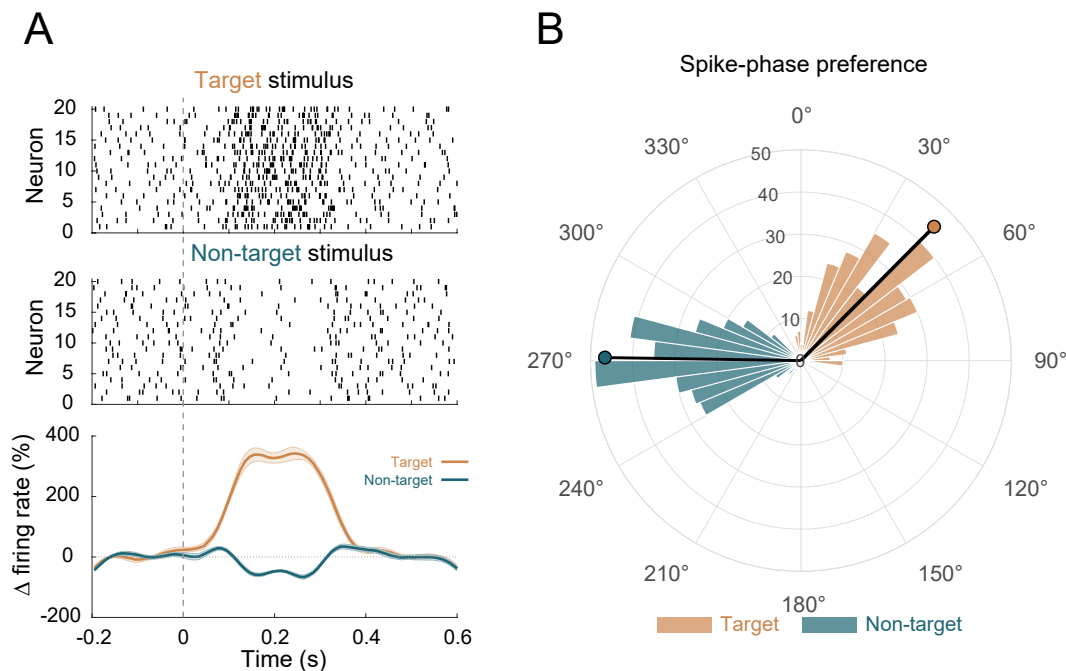


Figure A1: Appendix A example figure. (A) Shows raster plots of single neuron activity relative to the onset of target (orange) and non-target (teal) stimuli, followed by a peri-stimulus time histogram (PSTH) of their trial averages. (B) Polar histogram of preferred local field potential phases for recorded neurons following target and non-target stimuli.

A.3 Supplemental tables

B Something Else

Another appendix!